(a) Bounded-width dynamic routing for RL

RW32: fixed buffer width, variable active expert count

Default: $W_{\text{route}} = 256$

Reserved expert slots

RW32: $W_{\text{route}} = 32$

32 1/8 of reserved slots

$k_t^{\text{cap}} = |\mathcal{E}_t^{\text{cap}}|$

$k_t^{\text{cap}} \le 32$: preserve the expert set

$k_t^{\text{cap}} > 32$: retain top-scoring 32

Profile: P99 ≈ 19; > 32: 0.21%

(b) Final-route replay and asynchronous overlap

Final-route replay (R3)

Rollout
Final routes after capacity + RW32

Save routes

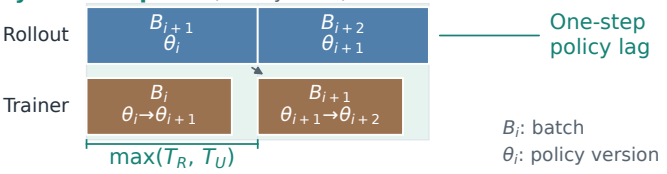
Actor replay
Reuse expert assignments
Recompute routing weights

Replay the final discrete expert path.

Synchronous



AsyncOverlap (steady state)



Idealized steady-state timing; overhead omitted.

(c) Capacity-constrained Quantile Routing



Inherited routing

Blue: active assignment

Cross: filtered assignment

Illustration: 4 tokens, 4 experts

Per-expert capacity = 2